PRODUCT USER MANUAL

For "Icebergs Observed with Sentinel-1 & RCM Number Density and Individual icebergs"

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007

Issue: 2.14

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.1	2011/05/27		Available time series is from April 1st 2010, and NOT 2011 as stated in the previous version (1.0) of the PUM.	Jørgen Buus-Hinkler	Jørgen Buus-Hinkler
1.2	2011/11/02	Layo ut	Conversion to new MyO PUM format	Jørgen Buus-Hinkler	Jørgen Buus-Hinkler
2.0	2011/11/17		Update file Nomenclature	L. Crosnier WP17	
2.1	2014/05/15	II	Add MFTP download mechanism	Bruce Hackett	Lars-Anders Breivik L. Crosnier
2.3	2015/01/30	III	Transition to Sentinel-1,SUBS, DGF Nomenclature	Jørgen Buus-Hinkler Bruce Hackett	L. Crosnier
2.4	2015/05/01	All	Change format to fit CMEMS graphical rules		L. Crosnier
2.5	2016/01/21	All	Finished CMEMS rebranding and content update for V2	Jørgen Buus-Hinkler	B. Hackett
2.6	2016/09/30		Inclusion of Sentinel-1B data in production, text updated accordingly	Jørgen Buus-Hinkler	B. Hackett
2.7	2018/01/25		Rebranded from OSI to SI TAC	C. Wettre	CMEMS products team
2.8	2019/04/03		Inclusion of Sentinel-1 IW mode for target detection, text updated accordingly	Jørgen Buus-Hinkler	C. Derval
2.9	2020/03/27		New CMEMS iceberg dataset (Individual Icebergs) in ESRI Shape-format is introduced.	Jørgen Buus-Hinkler	C. Wettre
2.10	2021/01/15	I.2, III.2, III.3	Update of statistical base: The base is now continuously updated and based on seasonal data. Previously it was generalized from all data. Moreover, statistical variables will start to be updated also in IW-netCDF files.	J. Buus-Hinkler	C. Wettre
2.11	2023/02/28	III.2	Added missing information about the presence of statistical variables in IW-mosaic-netCDF files in description table.	J. Buus-Hinkler	C. Wettre
2.12	2023/03/16	I.2, III.3	Changed info about AIS-based ship removal in history of change and in the description table for point based observations	J. Buus-Hinkler	C. Wettre
2.13	2024/05/28		Added figure 3c displaying geographical extension of product area of interest for point based observations towards pan-Arctic coverage	J. Buus-Hinkler	C. Wettre
2.14	2025/05/28		All products (now also gridded products in netCDF files) updated to pan-Arctic coverage: New figure 1 added, old figures 3a-b-c removed. Radarsat Constellation Imagery is now included in production, Main text, and dataset description are changed accordingly. PUM is now updated to latest CMEMS layout	J. Buus-Hinkler	C. Wettre

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GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
CFAR	Constant False Alarm Rate
DMI	Danish Meteorological Institute
DIDA	DMI Iceberg Detection Algorithm
DMI-ASIP	DMI Automated Sea Ice Products
NetCDF	Network Common Data Form
DU	Dissemination Unit
PU	Production Unit
AIS	Automatic Identification System, a maritime safety system that ships use to automatically inform about their location, course, and other data.
RCM	Radarsat Constellation Mission
SAR	Synthetic Aperture Radar
EW	Extra Wide Swath (Sentinel-1 acquisition mode)
IW	Interferometric Wide Swath (Sentinel-1 acquisition mode)
RCMLN	RCM Low Noise (RCM acquisition mode)
SI	Sea ice
GIS	Geographic Information System
SIGRID-3	SIGRID-3 is a vector archive format for Sea Ice Georeferenced Information and Data. It is a based on a WMO ice chart archive format first developed in 1981 and formalized in 1989, and its successor SIGRID-2. The vector format SIGRID-3, joins SIGRID and SIGRID-2 as standard WMO formats. The SIGRID-3 standard also includes codes for iceberg information.
TAC	Thematic Assembly Center

NRT	Near Real Time
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
MDS	Marine Data Store

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This guide describes the Copernicus Marine product: "Icebergs Observed with Sentinel-1 & RCM – Number Density and Individual Icebergs" (SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007) from the Copernicus Marine Sea Ice Thematic Assembly Centre (SI TAC), what data services are available to access them, and how to use the files and services. This product is produced at the Danish Meteorological Institute (DMI) (PU) and disseminated by the Copernicus Marine Data Store (MDS) via the Norwegian Meteorological Institute (MET Norway) (PC). The data are freely available under the Copernicus Marine Service data license.

The products are based on target (iceberg) detection in data inferred from SAR imagery (Sentinel-1 & RCM) processed using DIDA at the Danish Meteorological institute (DMI). The products only include icebergs detected in open-water areas (i.e. areas with no sea ice coverage) and include nine datasets based on Sentinel-1- and RCM SAR-scenes acquired in EW, IW, and RCMLN mode, respectively. The nine datasets consist of six sets of gridded "iceberg number density" available in netCDF format and three sets of "individual iceberg positions" available in ESRI Shapefile format:

1.	single scenes (EW)	**[gridded 10x10 km, netCDF format]**
2.	4-day mosaics (EW)	**[gridded 10x10 km, netCDF format]**
3.	single scenes (RCMLN)	**[gridded 10x10 km, netCDF format]** (new Nov-2025)
4.	4-day mosaics (RCMLN)	**[gridded 10x10 km, netCDF format]** (new Nov-2025)
5.	single scenes (IW)	**[gridded 10x10 km, netCDF format]**
6.	7-day mosaics (IW)	**[gridded 10x10 km, netCDF format]**
7.	4-day mosaics (EW)	**[point positions, Shapefile format]**
8.	4-day mosaics (RCMLN)	**[point positions, Shapefile format]** (new Nov-2025)
9.	7-day mosaics (IW)	**[point positions, Shapefile format]**

*Note: In this document, the terms **iceberg number density** and **iceberg concentration** are used interchangeably. Both refer to the number of icebergs detected within a 10×10 km grid cell.*

EW-scenes have a spatial coverage of approximately 400x400 km and the pixel spacing in the imagery is 40 m (true nominal resolution is 60-80 m). IW-scenes cover approximately 250x250 km – the pixel spacing in the imagery is 10 m (true nominal resolution is 80-30 m). RCMLN scenes cover approximately 350x350 km – the pixel spacing in the imagery is 40 m (true nominal resolution is ≈100 m)
The CMEMS iceberg products are solely based on dual polarized (HH, HV) SAR imagery.

b) History of Changes

- The first version of the iceberg number-density product became operational on April 02, 2010 and was based on SAR imagery from the Canadian Radarsat satellites.
- Since October 2014, Sentinel-1 data has gradually been incorporated into the product and was implemented operationally by February 13, 2015 (Today SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007 is solely based on Sentinel-1 SAR imagery).
- By February 15, 2016 the product was extended to include 4-day mosaics of iceberg number densities also including satellite acquisition time for each gridded observation. Important to note is that a new sea ice masking scheme was introduced for the mosaic-product. The scheme is based on high resolution sea ice charts. Since April 2017 this scheme has replaced the former AMSR-based ice-masking scheme for all iceberg products in the CMEMS product portfolio., 11
- In September 2016 Sentinel-1B data was implemented in the product line.
- From July 2019 Interferometric Wide (IW) imagery has been included in the production.
- Entry into Service for the first version of the new datasets in ESRI Shapefile format "Individual Icebergs" is 7 July 2020. The applied iceberg detection for these datasets is the same as for the existing "Number Density" datasets. The "Individual Icebergs" datasets are however substantially different and contain much more information than "Number Density"-datasets. Instead of number densities (i.e. number of icebergs per 10km x 10km) the geographical position of each individual detected iceberg is available together with information on iceberg size, detection significance, distance to nearest other iceberg, and distance to the sea ice edge plus more.
- With entry into service in May 2021, the statistical base and its derivation-scheme are changed (from general to seasonal). The update is made such that all observations acquired with the newest version of the detection algorithm (i.e. years 2017 and onwards) are automatically included whenever statistics is derived. The old "general statistical scheme" is replaced by seasonal statistics.
- With entry into service November 2023, the filtering of false targets is improved by introducing a new automated sea-ice filter based on DMIs Automated Sea Ice Products (ASIP). Furthermore, ships (misjudged as icebergs) are automatically removed by combining SAR satellite imagery acquisition time/spatial coverage with AIS ship tracking data.
- With entry into service November 2024 the geographical coverage of the product area of interest for the point (vector) based observations are extended towards pan-Arctic coverage.
- With entry into service in November 2025, the geographical coverage of the product area of interest for the gridded observations [netCDF] is expanded to include pan-Arctic coverage. In addition, three new datasets derived from RCM SAR imagery are introduced: single scene [netCDF], 4-day mosaics [netCDF], and point-based observations [shapefiles]. The geographical expansion of the gridded products necessitates a shift in the grid projection to the polar stereographic system (previously Lambert Conformal was used).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

The area in which iceberg detection is performed is indicated by blue in Figure 1. The whole grid provided in "iceberg concentration netCDF files" encompasses the entire Figure 1 and has a spatial resolution of 10 km. The spatial coverage- and the polar stereographic projection (EPSG 3411) is equivalent to what is used in the Arctic DMI-ASIP products: *SEAICE_ARC_PHY_AUTO_L3_MYNRT_011* _**023**- and _**024**

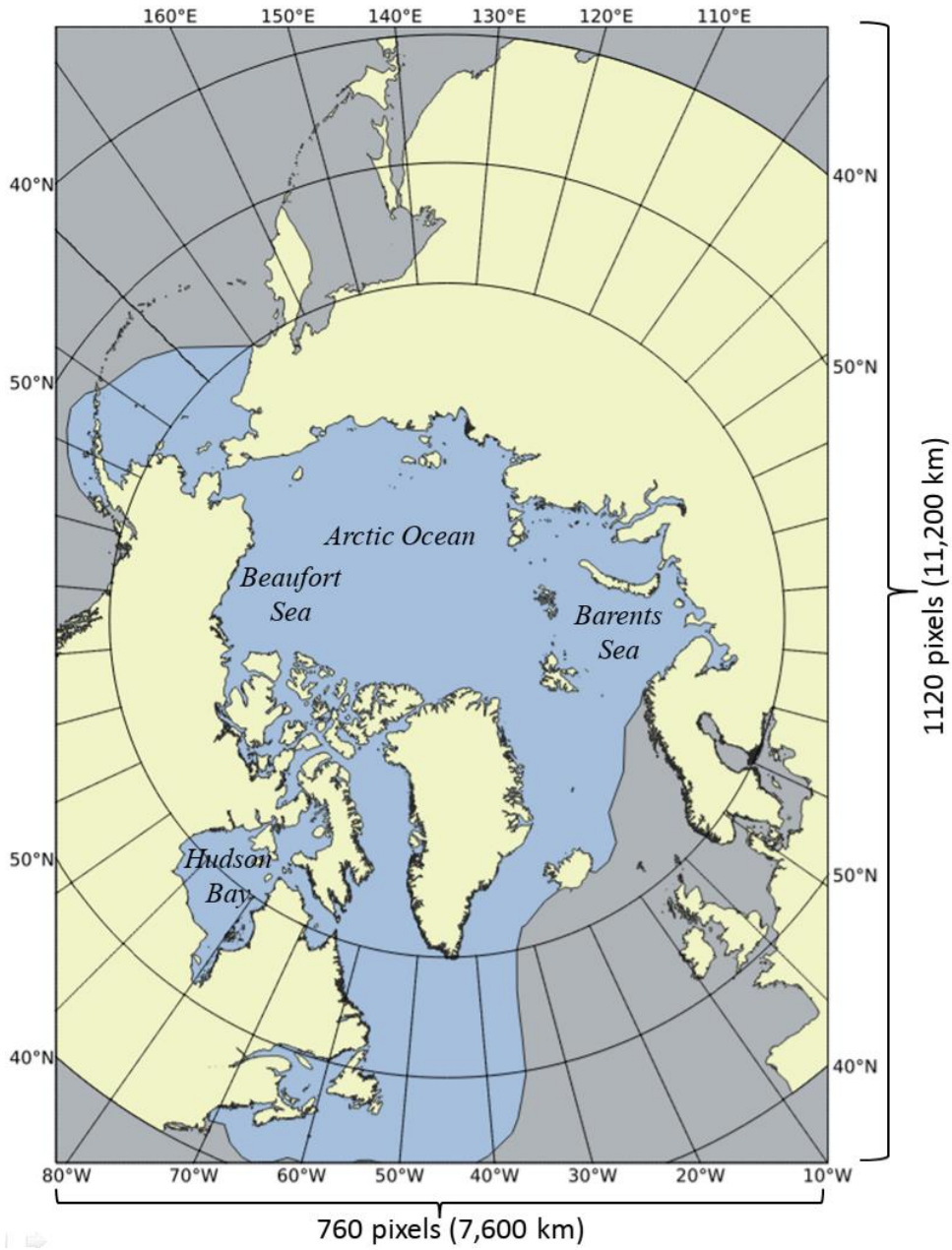

Figure 1. Iceberg detection area (blue) within the 10 km resolution iceberg-concentration-grid used in the netCDF files, matching the spatial coverage and projection (EPSG:3411) of DMI-ASIP Arctic products (which have spatial resolutions of 500-1000 m).

Number density *– six datasets in netCDF format*

Product Lines	SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007
Geographical coverage	Pan-Arctic
Variables	**Single scene products – three datasets (EW, IW, RCMLN):** Iceberg concentration (number density); Extremity index; Percentile corresponding to derived Extremity **4-day Mosaics – two datasets (EW, RCMLN):** Iceberg concentration (number density); Acquisition time for the individual grid cells; Extremity index; Percentile corresponding to derived Extremity **7-day Mosaic – one dataset (IW):** Iceberg concentration (number density); Acquisition time for the individual grid cells; Extremity index; Percentile corresponding to derived Extremity
Product Type	Near Real Time – Observations
Available time series	-2Y
Temporal resolution	4-day and 7-day mosaics are updated daily
Update Frequency	Continuously processed and sent to DU as close to NRT as possible.
Target delivery time	Deadline is within 24h from data acquisition (latest data acquisition for mosaics).
Horizontal resolution	10 km
Number of vertical levels	1
Grid	Polar stereographic 10 km resolution
Format	NetCDF4

Individual Icebergs *– three datasets in ESRI Shapefile format*

Product Lines	SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_007
Geographical coverage	Pan-Arctic
Variables – i.e. attributes in iceberg point Shapefile **In satellite coverage polygon Shapefile** **Note: Attributes are the same in both EW 4 day, and IW 7 day mosaics**	Attributes in SIGRID-3 standard: **RECDAT, IA_BLN, BRGARE, IA_BCN, POINT_TYPE** Attributes in Non-SIGRID-3 standard **Lon, lat, t_seconds, ib_dist_nm, ib_dist_km, icedist_km, cfar, sig_hh, sig_hv, ib_score, ais_chk, sat** **date_time, t_seconds, age_hours** (All Non-SIGRID-3) *All attributes are further described in section 2cin the Details of datasets table.*
Product Type	Near Real Time – Observations
Available time series	-2Y
Temporal resolution	Both the 4-day and 7-day IW mosaics are updated daily
Update Frequency	Continuously processed and sent to DU as close to NRT as possible.
Target delivery time	Continuously processed. Mosaics sent to DU once per day. Latest delivery within 24h from the latest data acquisition time in the given mosaic. One missing day is allowed for IW-mosaics Single scene-products are sent few hours after satellite acquisition. Latest delivery before 12 the following day.
Horizontal resolution	Not defined for point data
Number of vertical levels	1
Format	NetCDF4

b) Production System Description

The SAR instrument is an active sensor, which emits radar signals at an oblique angle towards the surface, and then measures the fraction of the signals that returns to the satellite (i.e. backscatter coefficient). Generally, obstacles (such as icebergs) will reflect a large fraction of the signal back towards the satellite (see Figure 2c), whereas a smooth surface (e.g. a calm sea) will act like a mirror reflecting virtually all the signal away from the satellite (Figure 2a).

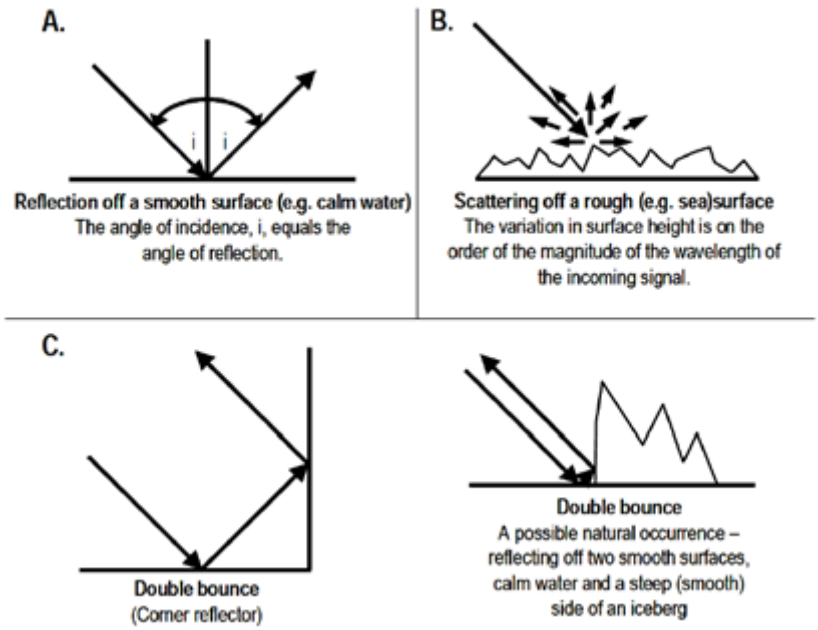

Figure 2. SAR scattering and reflection at different surface types. The typical reflection (double bounce) from large and steep targets, such as icebergs, is illustrated in the lower part (C) of the figure. Note that iceberg weathering (which leads to smoother icebergs with less sharp edges) and high winds (rugged sea surface) reduce the double bounce effect, which may make iceberg detection more difficult.

DIDA utilizes a Constant False Alarm Rate (CFAR) concept (followed by post processing routines). Basically this means that if a given pixel (backscatter value) is very unlikely to be part of the background (i.e. open sea) then this pixel will be classified as a target pixel. The procedure utilized by the CFAR algorithm is described as follows: Each pixel in the SAR imagery is investigated using a "sliding window" (see Figure 3). The pixel in question is the pixel in the centre of the window; the background is represented by the window's outer edge of pixels. The statistical distribution (Probability Density Function [PDF]) of the edge pixels is derived, and if the pixel in question is extremely unlikely to belong to the edge (i.e. background) distribution it will be classified as a target pixel.

After processing, the resulting CFAR image contains only none-iceberg- and iceberg pixels - the latter appearing as clusters made up of one or more pixels. To produce the final CMEMS products all iceberg pixel clusters are vectorized to iceberg target polygons – the position of a given individual iceberg target is defined as the centroid of its vectorized polygon. To derive the "Number Density"-products (netCDF files) all iceberg targets are counted in 10 km grid cells to produce an iceberg concentration, i.e. number density grid. The "individual iceberg positions" products (ESRI Shapefiles) contain automatically generated information about each individual iceberg target (such as iceberg size [length/area], geographical position, detection significance, local iceberg concentration, and more). This is further described under "Details of datasets" in section 2c.

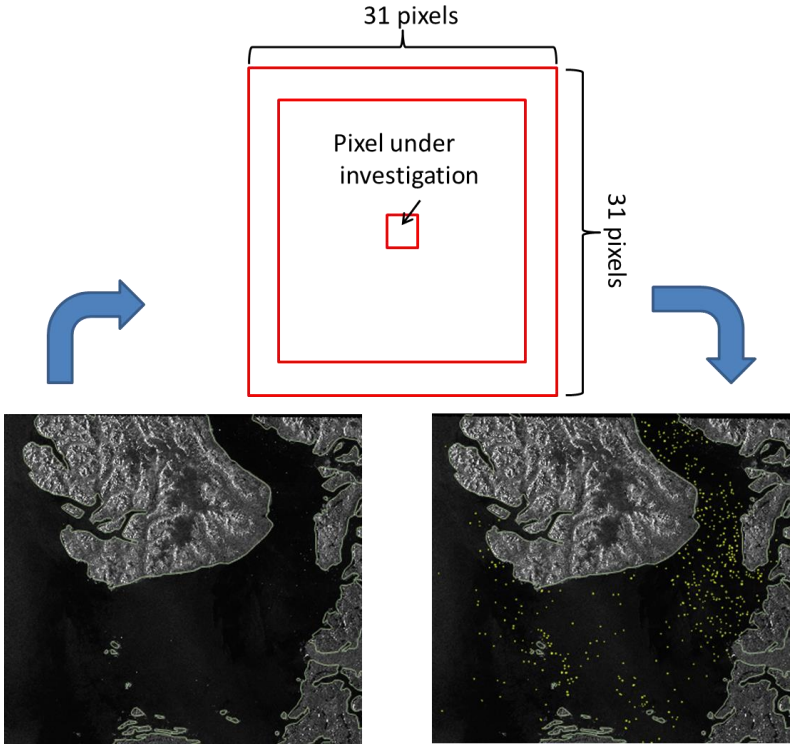

Figure 3. Conceptual description of CFAR target detection. The red part of the figure illustrates the moving window: every pixel in the SAR imagery (centre window pixel) is investigated by statistical comparison to the outer pixel ring (window edge one pixel wide). A probability density function (gamma distribution) is fitted to the histogram of the edge pixels (i.e. background pixels). If the value of the pixel in question is so high that it is very unlikely (less than a 10^{-7} chance) that it belongs to the background pixel distribution, then it is considered a target pixel. The bottom part of the figure shows an example of Sentinel-1 SAR imagery from Disko Bay, West Greenland: left: before iceberg detection, right: after iceberg detection with iceberg pixels highlighted in yellow.

Due to rugged structures in the sea ice (such as ridges and rubble fields) iceberg detection does not work reliably within sea ice infested areas and is thus only carried out in open water areas. To mask areas of sea ice, the DMI-ASIP Level-2 sea ice concentration products (available internally at DMI) are used. This approach is particularly well-suited because the DMI-ASIP products are geometrically aligned with the input SAR imagery used for iceberg detection. Both the iceberg and DMI-ASIP processing chains rely on the same SAR data, ensuring a perfect match in both spatial resolution and acquisition time.

This alignment eliminates spatial or temporal discrepancies between sea ice and iceberg detections. As a result, it significantly reduces the risk of false iceberg detections near the ice edge, where sea ice features can easily be misclassified as icebergs if not accurately masked. In addition to sea ice filtering, ship positions from an automated AIS dataflow at DMI are used to further reduce false detections. Iceberg candidates located near ship positions—within ±5 minutes of the satellite acquisition time—are automatically excluded from the dataset. For an illustration of the production flow with DIDA at DMI, see Figure 4.

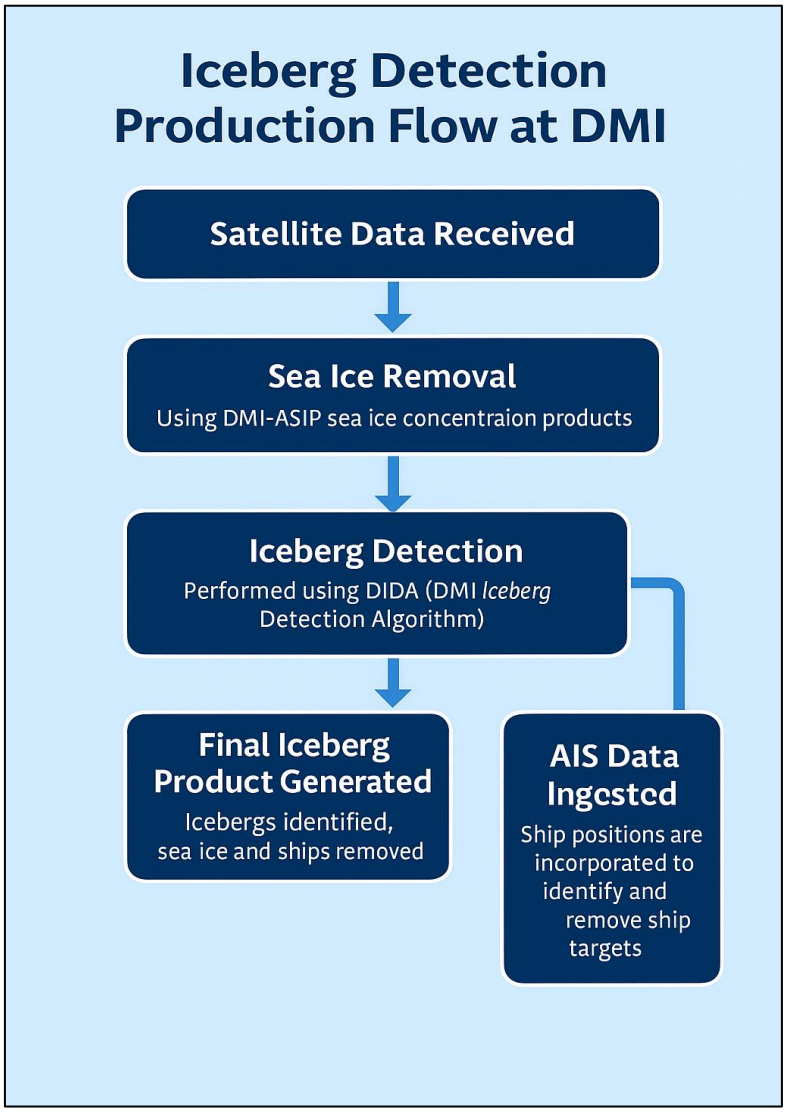

Figure 4. System chart for the DIDA production flow at DMI.

c) Dataset details

Table for **single-scene** number density **netCDF files**
(Note: three datasets in one table)

Dataset names (EW, IW, RCMLN)	DMI-ARC-SEAICE_BERG-L4-NRT-OBS DMI-ARC-SEAICE_BERG_IW-L4-NRT-OBS cmems_obs-si_arc_phy_berg-rcmln_nrt_l3-10km_irr
1. polar_stereo	Projection info
2. ibc	Iceberg concentration (number of icebergs per 100 km2, NODATA value: -1), data type: short integer.
3. extrem_lev	Iceberg number density level (L) (1=Normal [L<=P84], 2=Moderate [P84<L<=P97], 3=Extreme [>P97]) {always 1 when no icebergs are present}, data type: short integer.
4. stat_perc	Statistical percentile that corresponds to the current iceberg number density level within grid-cell in question, data type: short integer.
5. lat	Latitude (decimal degrees), data type: 32 bit float.
6. lon	Latitude (decimal degrees), data type: 32 bit float.
7. time	Days since 1970-01-01 00:00:00.
8. x	projection_x_coordinate, data type: double.
9. y	projection_y_coordinate, data type: double.

Table for **4-day** and **7-day mosaic** number density **netCDF files**
(Note: three datasets in one table)

Dataset names (EW, IW, RCMLN)	DMI-ARC-SEAICE_BERG_MOSAIC-L4-NRT-OBS (4 days) DMI-ARC-SEAICE_BERG_MOSAIC_IW-L4-NRT-OBS cmems_obs-si_arc_phy_berg-mos-rcmln_nrt_l3-10km_P1D_irr
1. polar_stereo	Projection info
2. ibc	Iceberg concentration (number of icebergs per 100 km2, NODATA value: -1), data type: short integer.
3. extrem_lev	Iceberg number density level (L) (1=Normal [L<=P84], 2=Moderate [P84<L<=P97], 3=Extreme [>P97]) {always 1 when no icebergs are present}, data type: short integer.
4. stat_perc	Statistical percentile that corresponds to the current iceberg number density level within grid-cell in question, data type: short integer.
5. acq_time	Acquisition time (seconds since 1970-01-01 00:00:00), data type: double.
6. lat	Latitude (decimal degrees), data type: 32 bit float.
7. lon	Latitude (decimal degrees), data type: 32 bit float.
8. time	Days since 1970-01-01 00:00:00.
9. x	projection_x_coordinate, data type: double.
10. y	projection_y_coordinate, data type: double.

Table for Iceberg point **Shapefiles: 4-day** and **7-day** mosaic products
(Note: three datasets in one table)

NOTE:	**Both the *iceberg point Shapefile set* and the *satellite coverage Shapefile set* are contained in the same zip-file.**
Dataset names (EW, IW, RCMLN)	cmems_sat-si_arc_berg-point_nrt_ew_d cmems_sat-si_arc_berg-point_nrt_iw_d cmems_obs-si_arc_phy_berg-point-mos-rcmln_nrt_l3_P1D_irr
0. EPSG 4326 (not an attribute – info available in .prj-file)	Projection info: wkt-format from .prj-file: GEOGCS["GCS_WGS_1984",DATUM["D_WGS_1984",SPHEROID["WGS_1984",6378137,298.257223563]],PRIMEM["Greenwich",0],UNIT["Degree",0.017453292519943295]] proj4 string: +proj=longlat +datum=WGS84 +no_defs
1. RECDAT	Observation date/time (=satellite acquisition time)
2. IA_BLN	IA_BLN: Maximum horizontal iceberg length (meters) at the waterline level – this distance is derived as the maximum distance (edge to edge) in the polygon circumferencing the pixel-cluster that makes up the iceberg in the SAR imagery
3. BRGARE	The area of the iceberg (square meters). I.e. the summed area of the pixels making up the iceberg
4. IA_BCN	Local iceberg concentration based on distance to nearest iceberg in nautical miles (nm). Distance categories in SIGRID3 code format: 10: >45 nm between icebergs 20: 15 - 44 nm between icebergs 30: 10 - 14 nm between icebergs 40: 7 - 10 nm between icebergs 50: 5 - 6 nm between icebergs 60: 3 - 4 nm between icebergs 70: 1 - 2 nm between icebergs 80: 0.5 - 1.0 nm between icebergs 90: <0.5 nm between icebergs 99: Undetermined/Unknown
5. POINT_TYPE	Point object category e.g. ICEBRG (SIGRID3 code for iceberg).
6. lon	Longitude (decimal degrees)
7. lat	Latitude (decimal degrees)
8. t_seconds	Observation time in seconds after, 1970-01-01 00:00:00
9. ib_dist_nm	Distance to nearest other iceberg in nautical miles
10. ib_dist_km	Distance to nearest other iceberg in kilometers
11. ice_dist_km	Distance to the sea-ice edge
12. cfar	Constant False Alarm Rate: This is a key parameter in the CMEMS iceberg detection algorithm. Its value is fixed at 10^{-7}. All pixels in the SAR imagery are tested by the algorithm and it is determined whether the given pixel is a target pixel or not. When a pixel is under investigation it is compared to the nearby pixels (background pixels defined as edge-pixels in the moving window). The statistical distribution of the

	background pixels is calculated; and if the value of the pixel in question is so high (i.e. above the locally derived threshold) that there is less than a 10^{-7} chance that it belongs to the background pixel distribution, then it is considered a target pixel.
13. sig_hh	Target detection significance in HH polarization: Defined as the percentage that the maximum-pixel-intensity value of the given iceberg-target is above the locally derived threshold. The local threshold is derived with the cfar-technique as described above in 12.
14. sig_hv	Target detection significance in HV polarization: Defined as the percentage that the maximum-pixel-intensity value of the given iceberg-target is above the locally derived threshold. The local threshold is derived with the cfar-technique as described above in 12.
15. ib_score	Iceberg score: The higher this score the more likely it is that the target is a true iceberg-target. Naturally, if a target is detected both in the HH-and HV polarization-channels it is more likely to be a real target than if only detected in one of them. Thus, in broad terms the score is calculated as a weighted sum of sig_hh and sig_hv. The weighted sum is however also adjusted by the normalized balance between the local HH and HV backscatter intensity (this is however beyond the technical scope of this document and is not fully explained here).Targets with ib_score < 40 are considered as false targets, and are thus excluded. Note that targets with ratio IA_BLN^2/ BRGARE > 2.5 are also excluded; since their Length to width ratio is too large for icebergs (they are more likely to be ships).
16. ais_chk	Shows whether the observation is checked for being located in the neighborhood of a registered ship-track. If the target is on/close to a track, then it is most likely a ship rather than an iceberg. If AIS data were available for the applied satellite scene this field will be assigned the distance to the nearest ship in km, otherwise it will get the value "NA".
17. sat	Name of the used SAR satellite

Satellite coverage polygon Shapefiles

0. EPSG 4326 **(not an attribute – info available in .prj-file)**	Projection info: wkt-format from .prj-file: GEOGCS["GCS_WGS_1984",DATUM["D_WGS_1984",SPHEROID["WGS_1984",6378137,298.257223563]],PRIMEM["Greenwich",0],UNIT["Degree",0.017453292519943295]] proj4 string: +proj=longlat +datum=WGS84 +no_defs
1. date_time	Observation date/time (=satellite acquisition time)
2. t_seconds	Observation time in seconds *after* 1970-01-01 00:00:00
3. age hours	Observation time in hours *before* latest acquisition time in file – more convenient for display on maps

d) Catalogue updating

Iceberg detection products with sea ice removal based on DMI-ASIP and ship removal using AIS data are updated continuously. Single-scene products are delivered as close to near-real time as possible—typically within a few hours of satellite acquisition and automated sea ice detection. Mosaicked products are updated daily.

e) Product use and impact

The Copernicus iceberg products, developed and delivered by DMI, provide near-real-time detection of icebergs in the Arctic waters, based on Synthetic Aperture Radar (SAR) satellite data. The products are primarily designed to support maritime safety and operational planning in regions where iceberg presence poses a hazard to navigation, and for use in scientific research.
Examples of key users of the product:

- **Operational ice services** (e.g. national or regional ice centers) that incorporate iceberg data into broader sea-ice analyses and forecasts.
- **Shipping and offshore operators** navigating through or working in high-latitude waters, where uncharted icebergs can pose a safety risk.
- **Climate and polar researchers** monitoring iceberg calving trends and freshwater fluxes into the ocean.

Value and Uniqueness in the Copernicus Product Portfolio
The iceberg products provide high-resolution, object-based information that complements sea-ice products (such as DMI-ASIP or ice charts provided by the national ice services). The iceberg products both deliver gridded iceberg concentration and individual iceberg detections with attributes such as position, size, and confidence score. The products consider open-water areas only, and are thus particularly valuable in regions where icebergs drift independently of the sea-ice field.

3) FILE FORMATS

The products are stored using the NetCDF and ESRI Shapefile formats.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

The "Individual Icebergs" point datasets are stored using the ESRI Shapefile format:
Where possible the attribute fields are named in compliance with the SIGRID3 standard for sea-ice and iceberg information.
A Shapefile is a vector data storage format for storing the location, shape, and attributes of geographic features. It is stored as a set of related files and can contain one feature class (i.e. lines, points or polygons). Shapefiles can contain large features with a lot of associated data and can be used in virtually all GIS applications.

An ESRI Shapefile (really a file set) consists of a main file, an index file, a dBASE table, and a projection information file.

Example:
- Main file: icebergs.shp
- Index file: icebergs.shx
- dBase table: icebergs.dbf
- Projection information file: icebergs.prj

The main file is a direct access, variable-record-length file in which each record describes a shape with a list of its vertices. In the index file, each record contains the offset of the corresponding main file record from the beginning of the main file. The dBASE table contains feature attributes with one record per feature. The one-to-one relationship between geometry and attributes is based on record number. Attribute records in the dBASE file must be in the same order as records in the main file.
Note that the projection information file is not strictly mandatory – however, if not present the Shapefile will most likely not be projected and displayed correctly in a GIS.
More in-depth information on the structural organization (headers, byte order etc.) can be found at:
https://www.esri.com/library/whitepapers/pdfs/Shapefile.pdf
Note that although the Shapefile format is still widely used, it has a long history as it was invented more than 20 years ago in the 1990's. In the documentation which can be achieved from the above mentioned link it is stated that all file names adhere to the 8.3 naming convention. This is however no longer true and this information should be updated. Shapefiles nowadays do support long file names.

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

4.1 FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
DMI-ARC-SEAICE_BERG-L4-NRT-OBS	iceberg_conc_yyyymmddhhmm.nc	0.005
DMI-ARC-SEAICE_BERG_IW-L4-NRT-OBS	iceberg_conc_iw_yyyymmddhhmmss.nc	0.005
cmems_obs-si_arc_phy_berg-rcmln_nrt_l3-10km_irr	iceberg_conc_rcmln_yyyymmddhhmmss.nc	0.005
DMI-ARC-SEAICE_BERG_MOSAIC-L4-NRT-OBS	iceberg_conc_mosaic_yyyymmddhhmm-yyyymmddhhmm.nc	0.005
DMI-ARC-SEAICE_BERG_MOSAIC_IW-L4-NRT-OBS	iceberg_conc_iw_mosaic_yyyymmddhhmm-yyyymmddhhmm.nc	0.005
cmems_obs-si_arc_phy_berg-mos-rcmln_nrt_l3-10km_P1D_irr	iceberg_conc_rcmln_mosaic_yyyymmddhhmm-yyyymmddhhmm.nc	0.005
cmems_sat-si_arc_berg-point_nrt_ew_d	iceberg_point_ew_mosaic_yyyymmddhhmm-yyyymmddhhmm.zip	0.02
cmems_sat-si_arc_berg-point_nrt_iw_d	iceberg_point_iw_mosaic_yyyymmddhhmm-yyyymmddhhmm.zip	0.02
cmems_obs-si_arc_phy_berg-point-mos-rcmln_nrt_l3_P1D_irr	iceberg_point_rcmln_mosaic_yyyymmddhhmm - yyyymmddhhmm.zip	0.02

4.2 STRUCTURE OF NetCDF- AND ESRI SHAPE FILES

Examples of the header of output NetCDF- and ESRI Shape-files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SEAICE-QUID-011-007.pdf

Product user manual DMI-ASIP
https://documentation.marine.copernicus.eu/PUM/CMEMS-SI-PUM-011-023-024.pdf

Quality Information Document DMI-ASIP
https://documentation.marine.copernicus.eu/QUID/CMEMS-SEAICE-QUID-011-023-024.pdf

J. Karvonen, A. Gegiuc, T. Niskanen, A. Montonen, J. Buus-Hinkler and E. Rinne, "Iceberg Detection in Dual-Polarized C-Band SAR Imagery by Segmentation and Nonparametric CFAR (SnP-CFAR)," in IEEE Transactions on Geoscience and Remote Sensing, vol. 60, pp. 1-12, 2022, Art no. 4300812, doi:10.1109/TGRS.2021.3070312.

J. Buus-Hinkler, K. Qvistgaard and K. A. H. Krane, "Iceberg number density — Reaching a full picture of the Greenland waters," 2014 IEEE Geoscience and Remote Sensing Symposium, Quebec City, QC, Canada, 2014, pp. 270-273, doi: 10.1109/IGARSS.2014.6946409.

R.S. Gill, "Operational Detection of Sea Ice Edges and Icebergs Using SAR", Canadian Journal of Remote Sensing, special issue on Sea Ice and Icebergs, Vol. 27, no. 5, pp. 411-432, 2001.

6) ANNEX

Structure and semantic of netCDF iceberg number density maps files

Single-scene products:

```
netcdf file://iceberg_conc_202505242118.nc {
  dimensions:
    x = 760;
    y = 1120;
    time = 1;
  variables:
    double x(x=760);
      :standard_name = "projection_x_coordinate";
      :units = "m";
      :_ChunkSizes = 760U; // uint

    double y(y=1120);
      :standard_name = "projection_y_coordinate";
      :units = "m";
      :_ChunkSizes = 1120U; // uint

    float lon(y=1120, x=760);
      :standard_name = "longitude";
      :units = "degrees_east";
      :_ChunkSizes = 1120U, 760U; // uint

    float lat(y=1120, x=760);
      :standard_name = "latitude";
      :units = "degrees_north";
      :_ChunkSizes = 1120U, 760U; // uint

    int polar_stereo;
      :grid_mapping_name = "polar_stereographic";
      :standard_parallel = 70L; // long
      :false_easting = 0.0; // double
      :false_northing = 0.0; // double
      :semi_major_axis = 6378273.0; // double
      :semi_minor_axis = 6356889.449; // double
      :spatial_ref = "EPSG:3411";
      :proj4_string = "+proj=stere +lat_0=90 +lat_ts=70 +lon_0=-45 +x_0=0
+y_0=0 +a=6378273 +b=6356889.449 +units=m +no_defs";
      :straight_vertical_longitude_from_pole = -45L; // long
      :latitude_of_projection_origin = 90L; // long

    short ibc(time=1, y=1120, x=760);
      :_FillValue = -1S; // short
      :long_name = "Iceberg concentration (number of icebergs per 100 km2)";
      :standard_name = "number_of_icebergs_per_unit_area";
      :units = "1e-2 km-2";
      :grid_mapping = "polar_stereo";
      :_ChunkSizes = 1U, 1120U, 760U; // uint

    short extrem_lev(time=1, y=1120, x=760);
      :_FillValue = -1S; // short
      :long_name = "Iceberg number density level (L) (1=Normal [L<=P84,
2=Moderate[P84<L<=P97], 3=Extreme[>P97]) {allways 1 when no icebergs are
present}";
      :units = "1";
      :grid_mapping = "polar_stereo";
```

```
      :_ChunkSizes = 1U, 1120U, 760U; // uint

    short stat_perc(time=1, y=1120, x=760);
      :long_name = "Statistical percentile that corresponds to the current
iceberg number density level within grid-cell in question";
      :_FillValue = -1S; // short
      :units = "%";
      :grid_mapping = "polar_stereo";
      :_ChunkSizes = 1U, 1120U, 760U; // uint

    double time(time=1);
      :calendar = "standard";
      :standard_name = "time";
      :units = "days since 1970-1-1 0:0:0";

  // global attributes:
  :title = "Arctic Iceberg Density, L4, 10km daily  (DMI-ARC-SEAICE_BERG-L4-
NRT-OBS)";
  :Conventions = "CF-1.6";
  :netcdf_version_id = "4.1.1 of Jan  7 2014 15:23:50 $";
  :creation_date = "2025-05-25";
  :product_version = "1.0";
  :software_version = "2.0";
  :comment = "Product User Manual (PUM) can be accessed at
http://marine.copernicus.eu";
  :sensor = "Sentinel-1 SAR (Extra Wide Swath mode at 80 m effective spatial
resolution)";
  :spatial_resolution = "10.0 km";
  :area = "Arctic";
  :southernmost_latitude = "40";
  :northernmost_latitude = "90";
  :westernmost_longitude = "-180";
  :easternmost_longitude = "180";
  :field_type = "daily";
  :institution = "Danish Meteorological Institute (DMI)";
  :institution_references = "http://www.dmi.dk";
  :contact = "jbh@dmi.dk";
  :operational_status = "operational";
}
```

Mosaic products:

```
netcdf file:/dmidata/projects/asip-
cms/JBH/copernicus_iceberg_update_2025/new_nc_test_files/iceberg_conc_mosaic
_202505210706-202505250452.nc {
  dimensions:
    x = 760;
    y = 1120;
    time = 1;
  variables:
    double x(x=760);
      :standard_name = "projection_x_coordinate";
      :units = "m";
      :_ChunkSizes = 760U; // uint

    double y(y=1120);
      :standard_name = "projection_y_coordinate";
      :units = "m";
      :_ChunkSizes = 1120U; // uint

    float lon(y=1120, x=760);
      :standard_name = "longitude";
      :units = "degrees_east";
      :_ChunkSizes = 1120U, 760U; // uint

    float lat(y=1120, x=760);
      :standard_name = "latitude";
      :units = "degrees_north";
      :_ChunkSizes = 1120U, 760U; // uint

    int polar_stereo;
      :grid_mapping_name = "polar_stereographic";
      :standard_parallel = 70L; // long
      :false_easting = 0.0; // double
      :false_northing = 0.0; // double
      :semi_major_axis = 6378273.0; // double
      :semi_minor_axis = 6356889.449; // double
      :spatial_ref = "EPSG:3411";
      :proj4_string = "+proj=stere +lat_0=90 +lat_ts=70 +lon_0=-45 +x_0=0
+y_0=0 +a=6378273 +b=6356889.449 +units=m +no_defs";
      :straight_vertical_longitude_from_pole = -45L; // long
      :latitude_of_projection_origin = 90L; // long

    short ibc(time=1, y=1120, x=760);
      :_FillValue = -1S; // short
      :long_name = "Iceberg concentration (number of icebergs per 100 km2)";
      :standard_name = "number_of_icebergs_per_unit_area";
      :units = "1e-2 km-2";
      :grid_mapping = "polar_stereo";
      :_ChunkSizes = 1U, 1120U, 760U; // uint

    double acq_time(time=1, y=1120, x=760);
      :_FillValue = -1.0; // double
      :long_name = "Acquisition time (seconds since 1970-1-1 0:0:0)";
      :units = "s";
      :grid_mapping = "polar_stereo";
      :_ChunkSizes = 1U, 560U, 380U; // uint

    short extrem_lev(time=1, y=1120, x=760);
```

```
      :long_name = "Iceberg number density level (L) (1=Normal [L<=P84,
2=Moderate[P84<L<=P97], 3=Extreme[>P97]) {allways 1 when no icebergs are
present}";
      :_FillValue = -1S; // short
      :units = "1";
      :grid_mapping = "polar_stereo";
      :_ChunkSizes = 1U, 1120U, 760U; // uint

    short stat_perc(time=1, y=1120, x=760);
      :_FillValue = -1S; // short
      :long_name = "Statistical percentile that corresponds to the current
iceberg number density level within grid-cell in question";
      :units = "%";
      :grid_mapping = "polar_stereo";
      :_ChunkSizes = 1U, 1120U, 760U; // uint

    double time(time=1);
      :standard_name = "time";
      :units = "days since 1970-1-1 0:0:0";
      :calendar = "standard";

  // global attributes:
  :title = "Arctic Iceberg Concentration Mosaic, L4, 10km daily  (DMI-ARC-
SEAICE_BERG_MOSAIC-L4-NRT-OBS)";
  :Conventions = "CF-1.6";
  :netcdf_version_id = "4.1.1 of Jan  7 2014 15:23:50 $";
  :creation_date = "2025-05-25";
  :product_version = "1.0";
  :software_version = "2.0";
  :comment = "Product User Manual (PUM) can be accessed at
http://marine.copernicus.eu";
  :sensor = "Sentinel-1 SAR (Extra Wide Swath mode at 80 m effective spatial
resolution)";
  :spatial_resolution = "10.0 km";
  :area = "Arctic";
  :southernmost_latitude = "40";
  :northernmost_latitude = "90";
  :westernmost_longitude = "-180";
  :easternmost_longitude = "180";
  :field_type = "daily";
  :institution = "Danish Meteorological Institute (DMI)";
  :institution_references = "http://www.dmi.dk";
  :contact = "jbh@dmi.dk";
  :operational_status = "operational";
}
```

a) Structure and semantic of Shapefiles "Individual Icebergs"

4 day EW and 7 day IW mosaic products:

Shapefile (iceberg point positions) attribute field specifications:

Name	Type	Type name	Length	Precision
RECDAT	QString	String	20	0
IA_BLN	int	Integer	5	0
BRGARE	qlonglong	Integer64	10	0
IA_BCN	int	Integer	2	0
POINT_TYPE	QString	String	10	0
lon	double	Real	9	7
lat	double	Real	9	7
t_seconds	qlonglong	Integer64	10	0
ib_dist_nm	double	Real	9	7
ib_dist_km	double	Real	9	7
icedist_km	qlonglong	Integer64	10	0
cfar	double	Real	9	7
sig_hh	qlonglong	Integer64	10	0
sig_hv	qlonglong	Integer64	10	0
ib_score	qlonglong	Integer64	10	0
ais_chk	QString	String	10	0
sat	QString	String	10	0

Other info:
proj4_string = " +proj=longlat +datum=WGS84 +no_defs";
Institution = "Danish Meteorological Institute (DMI)";
institution_references = "http://www.dmi.dk";
contact = "jbh@dmi.dk";
operational_status = "operational"

Shapefile (satellite coverage) attribute field specifications:

Name	Type	Type name	Length	Precision
date_time	QString	String	80	0
t_seconds	QString	String	80	0
age_hours	int	Integer	9	0

Other info:
proj4_string = " +proj=longlat +datum=WGS84 +no_defs";
Institution = "Danish Meteorological Institute (DMI)";
institution_references = "http://www.dmi.dk";
contact = "jbh@dmi.dk";
operational_status = "operational"

Reading software netCDF files

NetCDF data can be browsed and used through a number of software, like:

1. ncBrowse: http://www.epic.noaa.gov/java/ncBrowse/
2. NetCDF Operator (NCO): http://nco.sourceforge.net/
3. IDL, Matlab, Python, GMT…

Reading software ESRI Shapefiles

ESRI Shapefiles can be browsed and used through a number of software, like:

4. Quantum GIS: https://qgis.org/en/site/
5. ArcGIS (ESRI software): https://www.esri.com/en-us/home
6. Global Mapper: https://www.bluemarblegeo.com/products/global-mapper.php
7. Python gdal, ogr, osr packages